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 Oefeningenreeks 6G nr. 2.1

$$f(x) = \begin{cases} 0 & \text{als } x \in [-\pi, 0[\\ 4 & \text{als } x \in [0, \pi[\\ 7 & \text{als } x \in \{-\pi, 0, \pi\} \end{cases}$$

$\Rightarrow f(1) \neq f(-1) \Rightarrow$ niet even

$\Rightarrow f(1) \neq -f(1) \Rightarrow$ niet oneven

$$\Rightarrow f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$\Rightarrow a_0 = \frac{1}{\pi} \left(\int_{-\pi}^0 0 dx + \int_0^{\pi} 4 dx \right) = \frac{1}{\pi} \left(\int_0^{\pi} 4 dx \right) = 4$$

$$\Rightarrow a_n = \frac{1}{\pi} \left(\int_0^{\pi} 4 \cos(nx) dx \right) = \frac{4}{\pi} \int_0^{\pi} \cos(nx) dx$$

$$= \frac{4}{\pi} \left[\frac{\sin(nx)}{n} \right]_0^{\pi} = \frac{4}{\pi} \cdot 0 = 0$$

$$\Rightarrow b_n = \frac{1}{\pi} \left(\int_0^{\pi} 4 \sin(nx) dx \right) = \frac{4}{\pi} \int_0^{\pi} \sin(nx) dx$$

$$= \frac{4}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{4}{\pi} \left[-\frac{\cos(n\pi) - \cos(0)}{n} \right] = \frac{4}{\pi} \left[-\frac{(-1)^n - 1}{n} \right]$$

$$\Rightarrow n \text{ even} \Rightarrow b_n = \frac{4}{\pi} \left(-\frac{0}{n} \right) = 0$$

$$\Rightarrow n \text{ oneven} \Rightarrow b_n = \frac{4}{\pi} \left(\frac{2}{n} \right) = \frac{8}{n\pi}$$

$$\Rightarrow f(x) = 2 + \sum_{n=1}^{\infty} \left(-4 \left(\frac{(-1)^n - 1}{n\pi} \right) \sin(nx) \right)$$

alleen over oneven getallen:

$$= 2 + \frac{8}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}$$

References